A3-PS003-TC011

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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AIORI Hackathon 3

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Problem Description

- Traffic-flow forecasting is central to smart-city signal control and planning.
- Centralised training pools raw sensor data in one place, raising privacy and bandwidth concerns.
- Federated learning (FedAvg) keeps data at each simulated intersection and shares only model updates.
- Open questions: how do **non-IID client data** and **communication overhead** affect accuracy and convergence?
- Goal: compare centralised vs federated forecasting on one public dataset across 3-5 simulated clients, over 3 random seeds.
- **Scope limits:** simulation only; no live cameras, no city-wide deployment, no formal privacy guarantees (we claim only that raw data stays on clients).

Background

- **Traffic-flow forecasting:** predict near-future flow/speed from recent sensor history (time-series regression).
- **FedAvg [1]:** clients train locally for a few epochs; the server averages their weights each round.
- **IID vs non-IID:** in practice each intersection sees a different traffic pattern, so client data is non-IID, which can slow or destabilise convergence [3].
- **Communication cost:** each round transfers model weights up and down; bytes per round and rounds-to-converge define the cost.
- **Evaluation hygiene:** chronological train/test splits prevent leakage from future data.

Literature & RFC Review

- **McMahan et al., 2017 [1]:** introduces FedAvg; our core algorithm.
- **Beutel et al., 2020 [2]:** Flower framework; we use it to simulate clients on one machine.
- **Li et al., 2020 [3]:** FedProx; analyses the effect of heterogeneous (non-IID) client data on FL.
- **Li et al., 2018 [4]:** DCRNN; introduces the METR-LA traffic benchmark and forecasting baselines.
- **TensorFlow Federated:** alternative FL framework named in the problem statement.

Proposed Solution

- **Data:** one public traffic dataset (planned: METR-LA), split across 3-5 simulated clients (intersections/sensor groups).
- **Model:** one compact GRU/LSTM forecaster, identical in centralised and federated runs.
- **Federated setup:** FedAvg via Flower, local epochs per round and number of rounds fixed per experiment.
- **Splits:** chronological train/test split; client partitions built in two ways: IID (random) and non-IID (by location).
- **Metrics:** MAE, RMSE, communication bytes per round, rounds to converge.
- **Reproducibility:** 3 random seeds, fixed configs, one-command scripts; mean and spread reported.
- **Extended analysis (our USP):** accuracy vs communication-budget curve, update compression, tunable non-IID severity, client dropout (see next slide).

Optimization Proposed

Aspect	Required baseline	Our extension (USP)	Measured by
Communication	Bytes per round	MAE vs cumulative bytes sent; bytes needed to reach a target MAE	Bytes-to-target, convergence
Update size	Full 32-bit updates	16-bit / 8-bit quantised (optional top-k) updates	Bytes saved vs MAE lost
Heterogeneity	IID vs non-IID (2 cases)	Tunable non-IID severity (e.g. Dirichlet skew), curve not two points	MAE gap vs skew level
Robustness	All clients every round	10-30% random client dropout per round	MAE and convergence change
Reliability	3 seeds	Fixed configs and one-command reproducible scripts	Mean and spread across seeds
Optional	FedAvg	FedProx as a third method for non-IID clients	MAE vs FedAvg

Presented as extended analysis, not a new algorithm. Results to be filled in after experiments.

Solution Architecture & Design

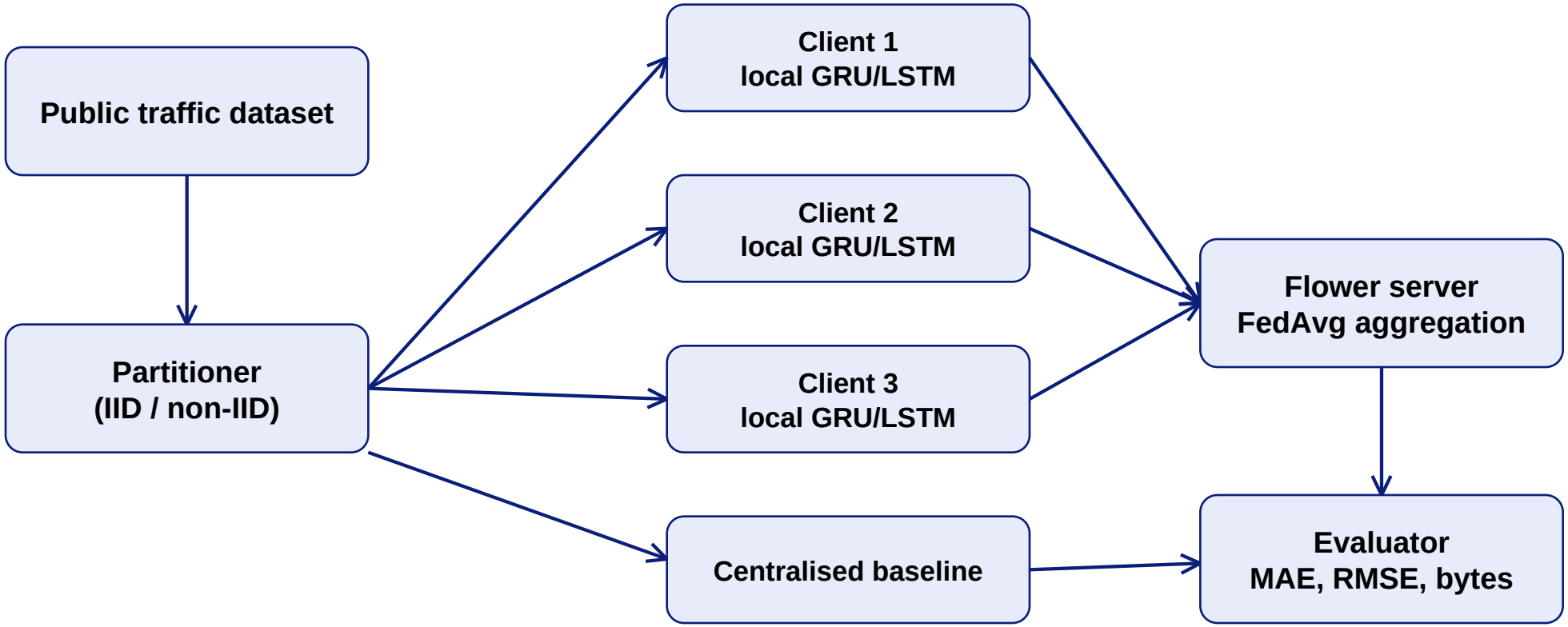
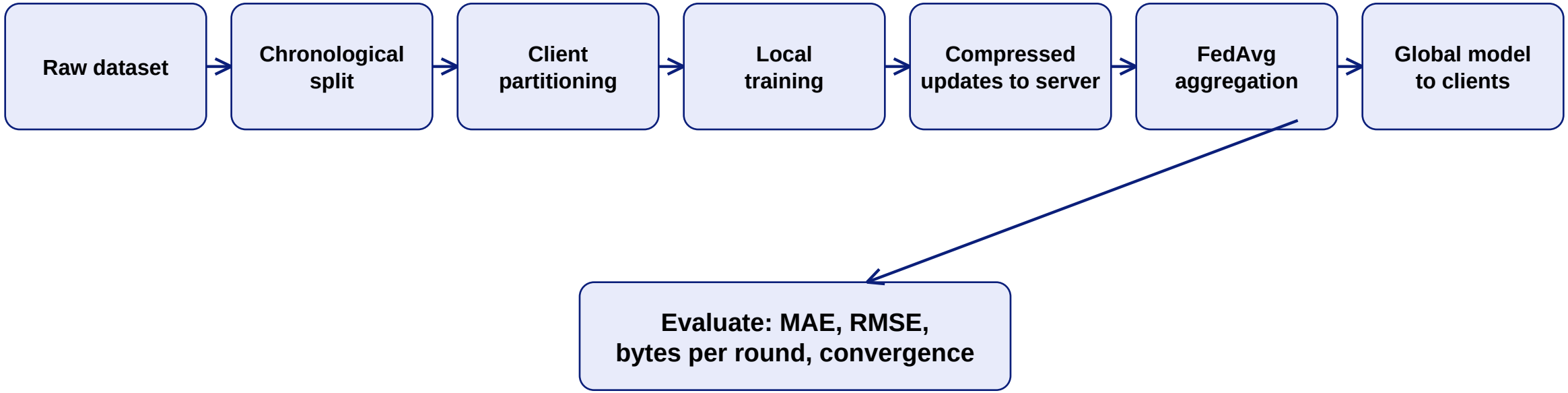


Figure: System architecture (clients, server, baseline, evaluation)

Data Flow Diagram



Steps 4-7 repeat each round; the loop runs for each partition type (IID, non-IID) and each seed.

Figure: Data flow diagram (DFD)

Deployment Timeline

- **Phase 1:** repository set up (A3-PS003-TC011), dataset download and preprocessing.
- **Phase 2:** centralised baseline and compact forecasting model.
- **Phase 3:** Flower FedAvg simulator with 3-5 clients.
- **Phase 4:** IID vs non-IID experiments, 3 seeds, communication-cost logging.
- **Phase 5:** USP experiments: bytes-to-target curves, compression, non-IID severity sweep, client dropout.
- **Phase 6:** analysis, plots, final report across 3 seeds, pseudocode, prototype demo to mentor.
- *Calendar dates to be aligned with the official hackathon schedule.*

Bibliography

- [1] H. B. McMahan et al. "Communication-Efficient Learning of Deep Networks from Decentralized Data." AISTATS, 2017.
- [2] D. J. Beutel et al. "Flower: A Friendly Federated Learning Framework." arXiv:2007.14390, 2020.
- [3] T. Li et al. "Federated Optimization in Heterogeneous Networks." MLSys, 2020.
- [4] Y. Li et al. "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting." ICLR, 2018.

Thank You for Your Attention!

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